A direct conditional comparison up to $ns = 1/50$

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Prepared with AI assistance. This is a conditional comparison theorem; it does not assert a complete Berry–Esseen proof.

1 Direct comparison on the full interval $0 < ns \le 1/50$

The following argument uses one outgoing count and elementary estimates for the incoming counts. Its only non-elementary probability input is the quadratic-hinge Poisson comparison stated below. It needs the original 3/5 conditional strip, but neither half-amplitude confinement nor fractional-part smoothing, marked crossings, or count curvature. The conclusion is uniform as the noise tends to zero.

Theorem 1. *Let $n \ge 42$, $2/5 \le p \le 53/100$, $q = 1 - p$, and $Y = B + U$, where $\mathbb{P}(B = 1) = p$ and $\mathbb{E}(U \mid B) = 0$. Write U_j for the conditional law of U given $B = j$. Assume*

$$|U_j| \le 3/5, \quad U_0 \le a_0, \quad U_1 \ge -a_1, \qquad a_0, a_1 \ge 0, \quad d := a_0 + a_1 \le 9/125.$$

Put $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $\ell = (v+s)^{-1/2}$, $X = \ell(Y-p)$, $\rho = \mathbb{E}|X|^3$, and $\rho_B = (1-2v)/\sqrt{v}$. Let S_n be the sum of n independent copies of X. Suppose

$$0 < \lambda \le U_* := 1/50, \qquad |H| \le 46/25.$$

Choose an integer k and a phase $u \in [-34/125, 91/125]$ with $np + H/\ell = k + u$, and put $\delta = k - np$. If F_B is the distribution function of $\mathrm{Bin}(n, p)$, then $1 \le k \le n - 1$ and, for every $C \ge 0$,

$$\begin{aligned}\Delta_C := \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < -\lambda/100.\end{aligned}$$

In particular, the sharp Bernoulli bound with $C = C_{\mathrm{E}}$ excludes a nonnegative fixed-target excess under these hypotheses.

The interval for u has length one, so the anchor always exists. The one-sided strip assumptions enter only the third-moment comparison. The comparison itself does not assume contact, a least counterexample, or a predecessor theorem. Those are separate obligations for acquiring its hypotheses in a global proof.

1.1 Conditioning and normalization

Let W_j be the sum of j independent variables with law U_1 and $n - j$ with law U_0, and let $\pi_j = \binom{n}{j} p^j q^{n-j}$, zero outside the legal counts. Define

$$f(u) = \frac{\Phi((\delta + u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k},$$

$$O = \mathbb{P}(W_k > u) + f(u), \qquad T = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u - i).$$

Conditioning gives the exact identity

$$\mathbb{P}(S_n \le H) - F_B(k) = -\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u+i) + \sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u-i).$$

Thus, if $\pi_{k+i}/\pi_k \le R_0$,

$$\Delta_C \le -\sqrt{n}\pi_k(O - R_0 T) + \mathcal{R}, \tag{1}$$

where

$$\mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

Outgoing events are strict and incoming events closed throughout.

Here $v \ge v_* := 6/25$ and $\sqrt{v+s} < 501/1000$. Write $\Xi = (46/25)(501/1000)$. Then $|\delta| < D := 33/20$ globally; if $|u| \le 1/20$, then $|\delta| < 1$. In particular k is interior and $9/25 < k/n < 23/40$. On the segment joining p to k/n,

$$|D'''(r\|p)| \le 6, \quad |(\log(r(1-r)))'| \le 5/4, \quad 1/(12r(1-r)) < 3/8.$$

The absolute values matter because this segment can cross $1/2$. Finite Stirling and Taylor's theorem therefore imply

$$\sqrt{n}\pi_k = \frac{\phi(0)}{\sqrt{v}} \exp\{-\delta^2/(2nv) + e\}, \qquad n|e| < Z(D_0) := 5D_0/8 + 3/8 + D_0^3/42 \tag{2}$$

whenever $|\delta| \le D_0 \le D$. This uses the finite formula $\log(j!) = (j+1/2)\log j - j + \frac{1}{2}\log(2\pi) + r_j$ with $0 < r_j < 1/(12j)$; its proof is retained in `appendices/count-comparison.tex`, at the paragraph beginning "For completeness, the finite Stirling formula used here is". The three errors in (2) are bounded by D_0^3/n^2, $5D_0/(8n)$, and $3/(8n)$.

For the derivative below, the actual count and δ are held fixed:

$$\log f'(t) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+t)^2}{2n(v+s)} - e.$$

If $|t| \le b$ and $|\delta| \le D_0$, this gives

$$n|\log f'(t)| \le \tfrac{8}{3}\lambda + \frac{2D_0 b + b^2}{2v_*} + Z(D_0). \tag{3}$$

Indeed $1/(2v_*) + D^2/(2 \cdot 42 v_*^2) < 8/3$ bounds the coefficient arising from the variance change. Using $0.39886 < \phi(0) < 0.4$, $e^{-x} \ge 1 - x$ and $e^x \le (1-x)^{-1}$ yields

	$\lvert u\rvert \le 1/20$	$-34/125 \le u \le 91/125$
$\sqrt{n}\pi_k$ lower bound	$73/100$	$13/20$
f' lower bound	$19/20$	$4/5$
f' upper bound	$21/20$	$5/4$
π_{k+i}/π_k upper bound	$53/50$	$3/2$
$\mathcal{R}$ upper bound	$2\lambda/n$	$3\lambda/n$

For the global ratio use $\sqrt{n}\max_j \pi_j < 17/20$, obtained from (2) at a mode, whose displacement is at most one. In the inner range the first exact binomial ratio is at most $1 + (53/100)/(42v_*) < 53/50$. Subsequent ratios are at most one because $(n+1)p - k < 153/100 < 2$.

Two improvements for positive phase will be used. For $0 \le u \le 1/10$ we have $|\delta| < 21/20$, and (3) gives $f'(t) > 19/20$. For $0 \le u \le 91/125$, use $\delta = \xi - u$, $|\xi| \le \Xi$. Since $t \mapsto 2\delta t + t^2$ is convex on $[0, u]$, its maximum is at most $\max(0, 2\Xi u - u^2)$. Thus

$$\log f'(t) \ge -\frac{1}{n}\left\{\frac{\lambda + 2\Xi(91/125) - (91/125)^2}{2v_*} + Z(D)\right\},$$

and $f'(t) > 9/10$. Here $91/125 < \Xi$, so $2\Xi u - u^2$ is increasing on the whole interval. No phase differentiation of the count is used.

To justify the repair row, $a_0 < p$ and $a_1 < q$ fix the two signs. The conditional cubic expansion, with $r = \min(p, q)$, gives

$$\mathbb{E}|Y - p|^3 \ge v(1 - 2v) + (3r - d)s.$$

Indeed $U_1^3 \ge -a_1 U_1^2$ and $U_0^3 \le a_0 U_0^2$. The inequality $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ then yields

$$(v + s)^{3/2}(\rho - \rho_B) \ge s\{3r - \tfrac{3}{2}(1 - 2v) - d - 3(1 - 2v)s/(8v)\} > 0.$$

The bracket is positive from $r \ge 2/5$, $v \ge v_*$, and $s \le 1/2100$. Hence the C term in $\mathcal{R}$ can be discarded for every $C \ge 0$. Differentiation of the normal term bounds it by $(7/4)\delta_+ s$, since $0.2/v_*^{3/2} < 7/4$. This proves the table.

Finally, if $s_j = \mathbb{E}U_j^2$, then $|s_1 - s_0| \le s/\min(p, q) \le (5/2)s$. Consequently

$$\frac{79}{84}\lambda \le \mathbb{E}W_k^2 \le \frac{89}{84}\lambda \quad (|u| \le 1/20), \qquad \mathbb{E}W_k^2 \le \frac{11}{10}\lambda \quad \text{globally}, \tag{4}$$

and $\mathbb{E}W_{k+i}^2 \le iV\lambda$ for $i \ge 1$, where $V = 9/8$ in the inner range and $V = 7/6$ globally.

1.2 A rational Poisson bound for all incoming counts

We use Pinelis, *On the Bennett–Hoeffding inequality*, arXiv:0902.4058, Proposition 2.8 and Corollary 2.11, printed page 9. If independent centered summands are bounded above by $A > 0$ and their total variance is at most V_0, their sum Z satisfies, with $\mu = V_0/A^2$,

$$\mathbb{P}(Z \ge b) \le \frac{\mathbb{E}(P_\mu - j)_+^2}{(b/A + \mu - j)^2} \quad \text{whenever } j < b/A + \mu, \tag{5}$$

where P_μ is Poisson with mean μ. To see the substitution, apply the stated quadratic-hinge moment comparison to $(x - A(j - \mu))_+^2$ and then Markov's inequality. This also proves the closed-tail version directly. In particular the theorem permits variance inflation before choosing j; no unproved monotonicity in μ is needed. We apply it to $-W_{k+i}$, whose summands also have upper bound $A = 3/5$.

For $i = 1$, choose $j = 1$. Since

$$\mathbb{E}(P_\mu - 1)_+^2 = \mu^2 - \mu + 1 - e^{-\mu} \le \mu^2(1/2 + \mu/6),$$

the first incoming probability, when $u \le h < 1 - A$, is at most

$$Q(\mu, h) := \frac{\mu^2(1/2 + \mu/6)}{((1 - h)/A - 1 + \mu)^2}, \qquad \mu = V\lambda/A^2. \tag{6}$$

The exponential inequality follows from Taylor's theorem with its nonnegative fourth-order remainder.

For $i \geq 2$, choose the integer hinge $j = i$ and inflate the variance to $iV\lambda$. If P is any nonnegative integer, then

$$(P-i)_+^2 \leq \binom{P}{i+1}, \qquad \mathbb{E}(P_{i\mu}-i)_+^2 \leq \frac{(i\mu)^{i+1}}{(i+1)!}.$$

For the first inequality write $P = i+r$, $r \geq 1$. At $i = 2$ it is $6r^2 \leq r(r+1)(r+2)$, equivalent to $(r-1)(r-2) \geq 0$ for integer r; increasing i increases the binomial coefficient. The expectation is the Poisson factorial moment, obtained directly by shifting its series. If $u \leq h$, with $h \geq 0$, set $b_h = 1/A - 1 - h/(2A) > 0$. Then

$$(i-u)/A + i\mu - i \geq i(b_h + \mu) > 0,$$

so all hinges are admissible and

$$\mathbb{P}(W_{k+i} \leq u - i) \leq \frac{i^{i-1}\mu^{i+1}}{(i+1)!(b_h+\mu)^2}.$$

The ratio of successive coefficients $i^{i-1}/(i+1)!$ is less than $e < 3$. The first term is $\mu^3/[3(b_h+\mu)^2]$. Therefore, when $3\mu < 1$,

$$\sum_{i\geq 2} \mathbb{P}(W_{k+i} \leq u - i) \leq J(\mu, h) := \frac{\mu^3}{3(b_h+\mu)^2(1-3\mu)}. \tag{7}$$

This is an infinite-tail bound, not a truncated numerical sum. Illegal counts are absent from the original sum and only increase its upper bound.

We will use $Q + J$ for negative and inner phases, and Cantelli plus J for positive outer phases. To obtain bounds relative to arbitrarily small λ, note that $J(\mu, h)/\lambda$ increases with μ: apart from a positive constant it is $\mu^2/\{(b_h+\mu)^2(1-3\mu)\}$. If $a_h = (1-h)/A - 1 > 0$ and $\mu \leq a_h$, then $Q(\mu, h)/\lambda$ also increases, being proportional to $\mu(a_h+\mu)^{-2}(1/2+\mu/6)$. All these conditions are checked below at $\lambda = U_*$.

1.3 A lower bound for the own count

For a centered independent sum W with summands bounded in absolute value by A and variance t, the moment expansion gives $|\mathbb{E}W^3| \leq At$ and $\mathbb{E}W^4 \leq A^2t + 3t^2$. Set $\alpha = 1/2$ and

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2x^2 + 2\alpha^3x}{5\alpha^4}.$$

The numerator factors as $-x(x+\alpha)^2(x-2\alpha)$, so it is nonpositive for $x \leq 0$; its positive maximum is $(9+6\sqrt{3})\alpha^4/4 < 5\alpha^4$. Thus $h_\alpha \leq \mathbf{1}_{(0,\infty)}$, including at zero. For $r \geq 0$ and $\sigma \in \{-1, 1\}$, expansion gives

$$\mathbb{P}(W + \sigma r > 0) \geq \frac{(3\alpha^2 - A^2)t - 3t^2 - 4Art - 6r^2t - r^4 + 3\alpha^2r^2 + 2\sigma\alpha^3r}{5\alpha^4}.$$

Put $T_* = (89/84)U_* = 89/4200$. After adding br, dividing by $t > 0$, and writing $r = zt$, the lower bound becomes

$$F_{\sigma,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{5\alpha^4} + \left(\frac{2\sigma}{5\alpha} + b - \frac{4At}{5\alpha^4}\right)z + \frac{3\alpha^2t - 6t^2}{5\alpha^4}z^2 - \frac{t^3z^4}{5\alpha^4}.$$

For fixed $z \geq 0$ this is concave in $t \geq 0$, so its minimum over $0 \leq t \leq T_*$ occurs at an endpoint. At each endpoint, replacing z^4 by $z_0^2 z^2$ gives a lower quadratic on $0 \leq z \leq z_0$. The endpoints and any eligible vertex give

$$\begin{aligned} &F_{1,-21/20}(z,t) > 3/8 && (0 \leq z \leq 2), \\ &F_{-1,19/20}(z,t) > 1 && (0 \leq z \leq 3/2). \end{aligned}$$

Their exact rational margins are at least respectively

$$\frac{72589549}{5788125000}, \qquad \frac{5518861753}{126554117375}.$$

This proves the bounds on rectangles, including their limiting edge $t = 0$; it is not a sampled small-noise assertion.

For $u = -r$ and $r \geq 2t$, Cantelli gives $O \geq r^2/(t + r^2) - (21/20)r$. At $r = 2t$ this is greater than $(3/8)t$, and its derivative in r is positive there. The derivative $2rt/(t + r^2)^2 - 21/20$ first increases and then decreases, because its derivative has the sign of $t - 3r^2$. Hence any later stationary point is a maximum, and the minimum on $[2t, 1/20]$ is at an endpoint. At $r = 1/20$ the desired bound is checked at $t = T_*$, since the left side decreases and $(3/8)t$ increases with t. The switch $2T_* < 1/20$ is valid. For positive phase the quartic handles $0 \leq r \leq 3t/2$; beyond it, $O \geq f(r) > (19/20)r > t$. Also $3T_*/2 < 1/20$. Together with (4), this proves

$$O > \frac{3}{8}\frac{79}{84}\lambda \quad \text{if } |u| \leq 1/20. \tag{8}$$

1.4 Four closed phase intervals complete the comparison

All remaining quantities in this paragraph are rational numbers evaluated at $U_* = 1/50$. Define

$$\mu_I = \frac{(9/8)U_*}{A^2} = \frac{1}{16}, \qquad \mu_G = \frac{(7/6)U_*}{A^2} = \frac{7}{108}.$$

The four values of b_h, for $h = 0, 1/20, 1/10, 91/125$, are $2/3, 5/8, 7/12, 3/50$, all positive; $3\mu_G < 1$. For the two uses of Q, $a_{1/20} = 7/12 > \mu_I$ and $a_0 = 2/3 > \mu_G$.

Inner phase: $|u| \leq 1/20$. Put $T_I = Q(\mu_I, 1/20) + J(\mu_I, 1/20)$. The monotonicity proved above gives $T \leq (T_I/U_*)\lambda$. The bracket in (1) is positive, and

$$\frac{73}{100}\left\{\frac{3}{8}\frac{79}{84} - \frac{53}{50}\frac{T_I}{U_*}\right\} - \frac{2}{42} - \frac{1}{100} = \frac{113307367}{16930513600} > 0.$$

This proves the strict comparison, uniformly for $\lambda \downarrow 0$.

Negative outer phase: $-34/125 \leq u \leq -1/20$. Put $T_G = (11/10)U_*$ and

$$L_- := \min_{r \in \{1/20, 34/125\}} \left\{\frac{r^2}{T_G + r^2} - \frac{5}{4}r\right\}.$$

The derivative is positive at $r = 1/20$ and has at most one later zero, a maximum, by the same derivative argument as above. Hence Cantelli gives $O \geq L_-$. Since $u \leq 0$, $T \leq Q(\mu_G, 0) + J(\mu_G, 0)$. The final sufficient margin is

$$\frac{13}{20}\left\{L_- - \frac{3}{2}[Q(\mu_G, 0) + J(\mu_G, 0)]\right\} - \frac{3U_*}{42} - \frac{U_*}{100} = \frac{95575595137}{4788968940000} > 0.$$

Positive near phase: $1/20 \le u \le 1/10$**.** Write $t_G = (7/6)U_*$. Then $O \ge (19/20)(1/20)$ and Cantelli bounds the first incoming term by $t_G/(t_G + (9/10)^2)$. Therefore

$$\frac{13}{20}\left\{\frac{19}{400} - \frac{3}{2}\left[\frac{t_G}{t_G + (9/10)^2} + J(\mu_G, 1/10)\right]\right\} - \frac{3U_*}{42} - \frac{U_*}{100} = \frac{8209}{4872000} > 0.$$

Positive far phase: $1/10 \le u \le 91/125$**.** Use $O \ge (9/10)u$. The function

$$u \longmapsto \frac{9}{10}u - \frac{3}{2}\frac{t_G}{t_G + (1-u)^2}$$

is concave on this interval, since $3(1-u)^2 > t_G$; its minimum L_+ is therefore attained at one of the two endpoints. Finally

$$\frac{13}{20}\{L_+ - \tfrac{3}{2}J(\mu_G, 91/125)\} - \frac{3U_*}{42} - \frac{U_*}{100} = \frac{389390249}{17290880250} > 0.$$

All brackets are positive before applying the lower mass bound. In the outer intervals, replacing λ by U_* only decreases the own-count lower bound and increases incoming tails and repairs. Thus these absolute margins imply the required comparison for every $0 < \lambda \le U_*$. The four closed intervals cover the full phase range, including their shared endpoints. This proves the theorem.

The 62 scalar gates in `moderate_noise_check.py` check all constants above by rational arithmetic. The analytic coverage follows from the monotonicity, concavity, and geometric-series arguments, not from a parameter grid. The theorem leaves $\lambda > 1/50$ and the remaining p panels open. In particular it does not justify deleting the existing half-amplitude or count certificates globally.